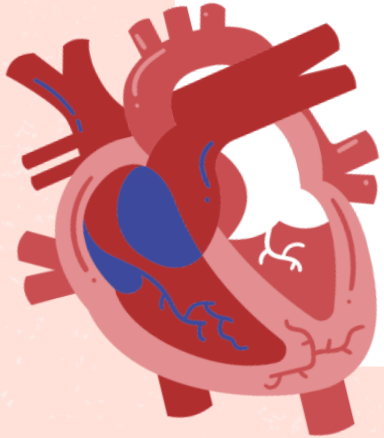
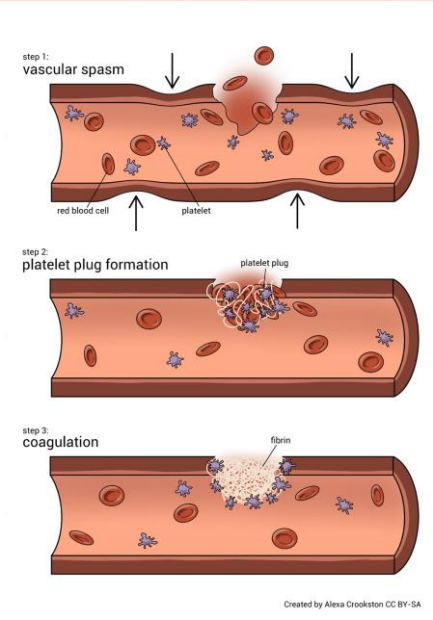
Disorders of Hemostasis



By: Ariana, Bri, Jordyn, Lizi, and Rachel

Three Stages of Hemostasis

1. Vessel Spasm
2. Formation of the platelet plug
3. Blood coagulation



Categories of Disorders of Hemostasis

- Thrombosis
  - The inappropriate formation of clots within the vascular system
- Bleeding
  - Failure of blood to clot in response to appropriate stimulus

Stage 1:Vascular Constriction

- Vessel spasm constricts the vessel with the intention of reducing blood flow

- Initiated by endothelial injury and caused by local and humoral mechanisms

Stage 2: Platelet Plug Formation

- Involves:
  - Activation, Adhesion, Aggregation
- Change from smooth disks to spiny spheres and expose glycoprotein receptors
- The protein ‘Willebrand factor’ allows platelets to stick to collagen fibers
- ADP is the mediator of platelet aggregation
  - Attracts more platelets to the site to form a clot

Intrinsic and Extrinsic Coagulation Pathways

- Two pathways in blood coagulation, but end result is the same
  - Calcium, factors X and V, and platelet phospholipids combine to form prothrombin activator
  - Prothrombin activator is then converted to thrombin
  - This interaction causes conversion of fibrinogen in fibrin strands that create the insoluble blood clot

Stage 3: Blood Coagulation

- Factor X, conversion of prothrombin to thrombin
- Thrombin is an enzyme that converts fibrinogen to fibrin, which then creates a meshwork
- This cements platelets and other components to form a clot
- Clot retraction after 20-60 min

Clot Retraction and Dissolution

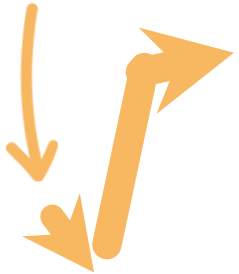
- Clot retraction and clot dissolution are also significant to hemostasis.

- The process involves the interaction of substrates, enzymes, protein cofactors, and calcium ions that circulate in the blood or are released from platelets and cells in the vessel wall.



Requirements for Blood Clotting Process

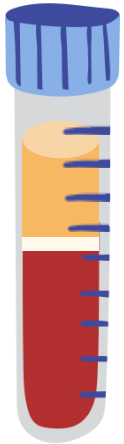
1. Presence of platelets produced in the bone marrow
2. Von Willebrand factor generated by the vessel endothelium
3. Clotting factors synthesized in the liver using vitamin K



Why Should Blood Clot?

- Hemostasis is designed to maintain the integrity of the vascular compartment
  - Infection
  - Volume
  - Oxygen production
  - Tissue damage
  - Healing

General Forms of Hypercoagulability States



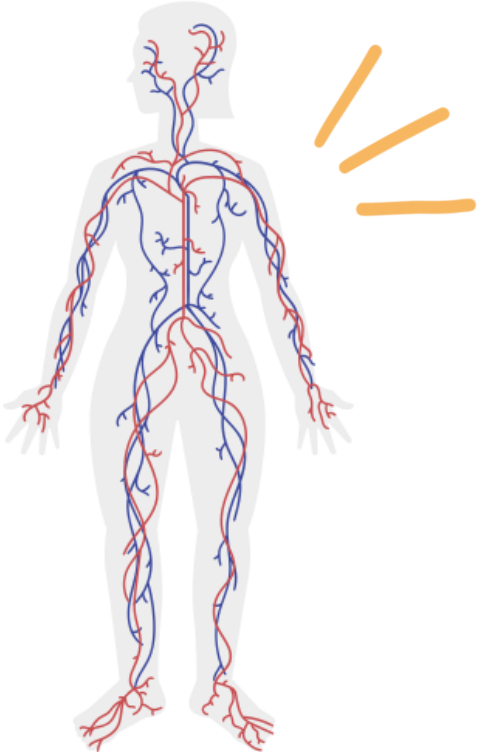
- Conditions that create increased platelet function:
  - Atherosclerosis
  - Diabetes mellitus
  - Smoking
- Elevated blood lipid and cholesterol levels
- Increased platelet levels

General Forms of Hypercoagulability States

- Conditions that cause accelerated activity of the coagulation system
  - Pregnancy and the puerperium
  - Use of oral contraceptives
  - Postsurgical state
  - Immobility
  - Congestive heart failure
  - Malignant diseases

Causes of Bleeding

- Platelet disorders
- Coagulation factors
- Blood vessel integrity



Warning signs

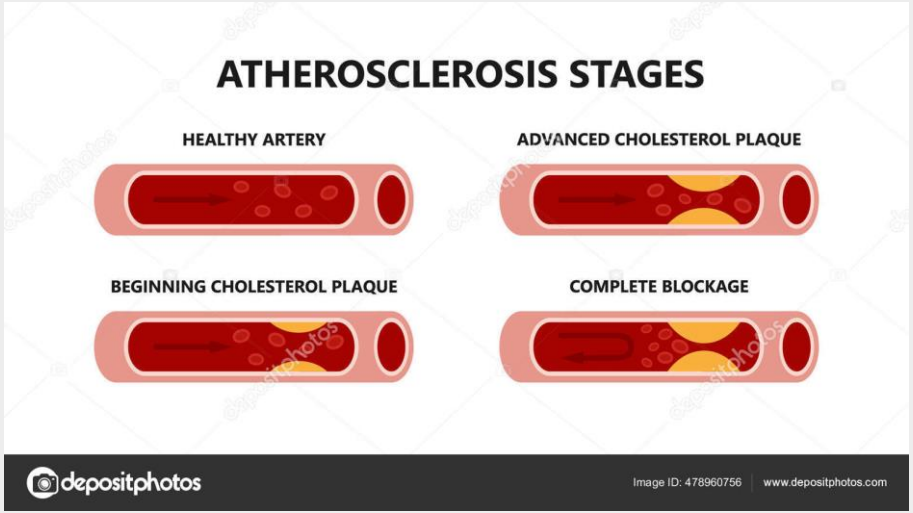
- Mucosal bleeding
- Purpura
- Petechiae

Increased Platelet Function

Hypercoagulability can cause our platelets to stick together which then causes clots and blocks blood flow.

The causes of increased platelet function are disturbances in flow, endothelial damage, and increased sensitivity of platelets to factors that cause adhesiveness and aggregation.

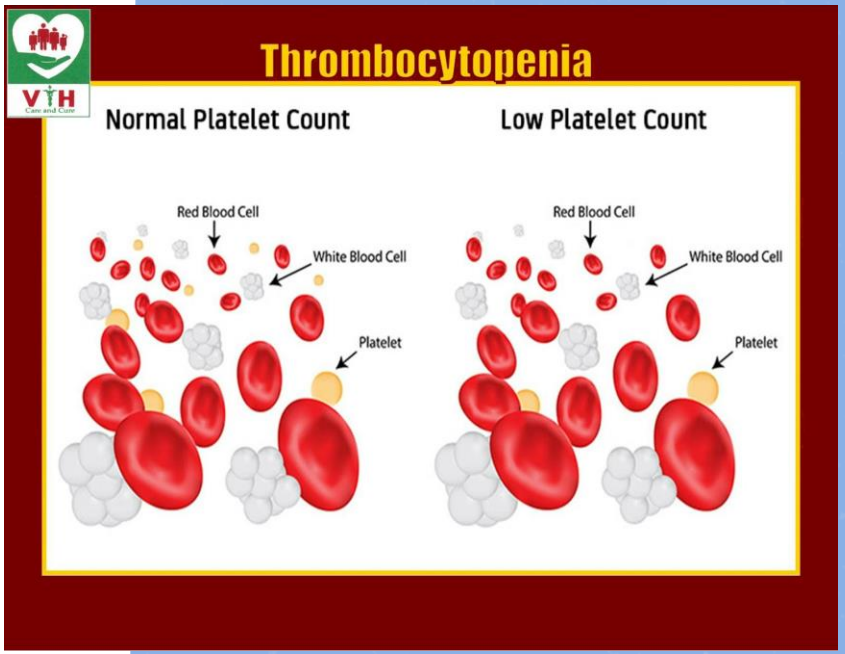
Atherosclerotic Plaques and Platelets



Platelet Defects

Thrombocytopenia

- Drug-induced thrombocytopenia
- Idiopathic thrombocytopenic purpura
- Thrombotic thrombocytopenic purpura



Coagulation and Vitamin K

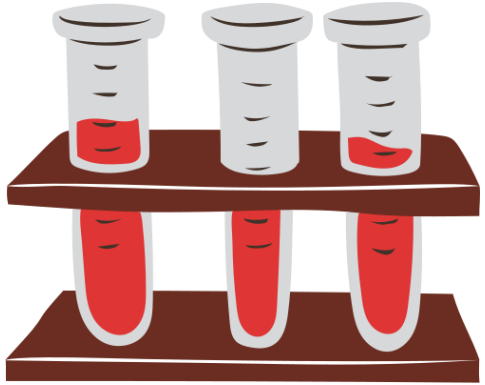
Blood

Vitamin K is an essential cofactor for synthesis of clotting factors.

Fat-soluble vitamin synthesized by intestinal bacteria

In vitamin K deficiency, the liver produces inactive clotting factor resulting in abnormal bleeding.

Coagulation Defects



- Deficiencies can arise because of defective synthesis, inherited disease, or increased consumption of the clotting factors
  - Hereditary disorders
    - Hemophilia A
    - Hemophilia B
    - Von Willebrand disease
    - Any genetic disruption of the production of clotting factor

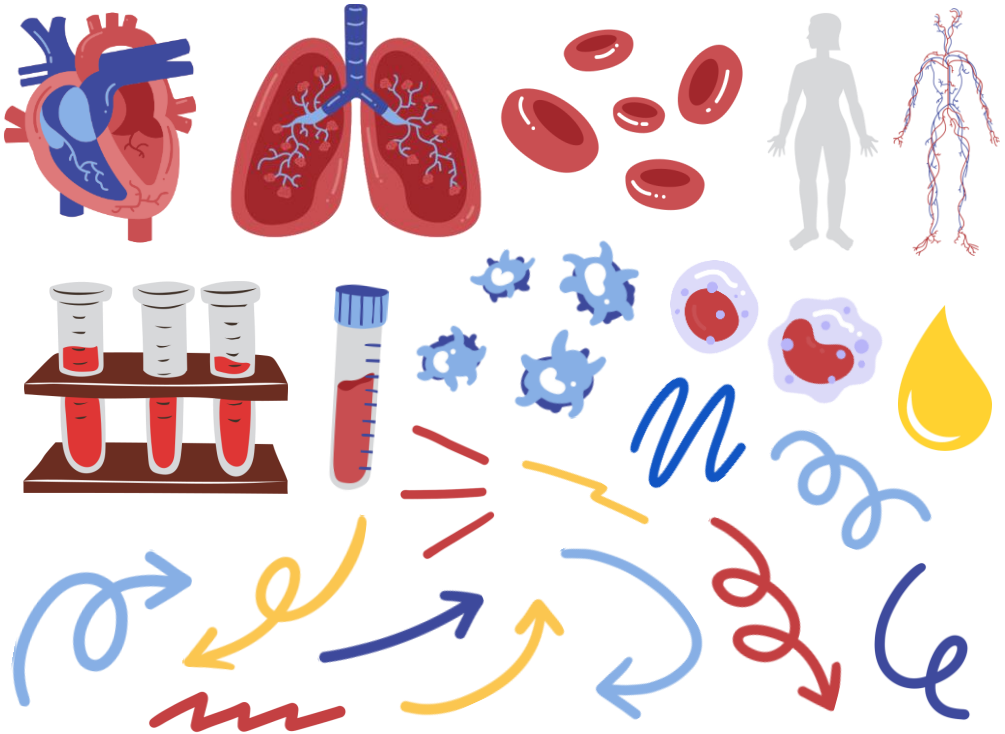
Vascular Disorders that Cause Bleeding #1

Hemorrhagic telangiectasia

Vitamin C Deficiency (scurvy)

Vascular Disorders that Cause Bleeding #2

- Cushing disease
- Senile purpura (bruising in elderly persons)



Conditions Associated with Disseminated Intravascular Coagulation

Obstetric conditions

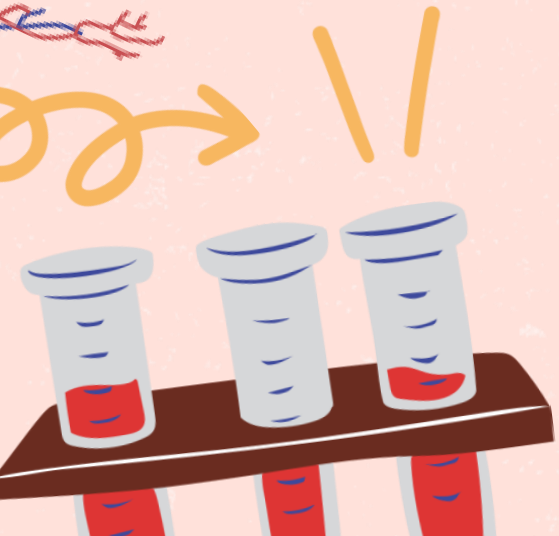
Cancers

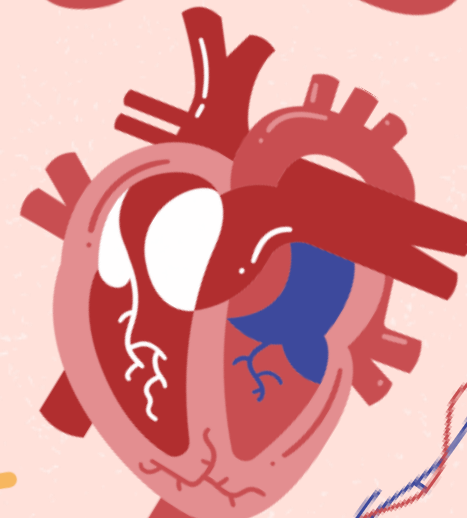
Infections

Shock

Trauma or surgery

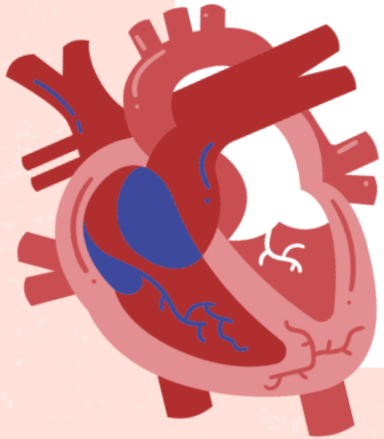
Hematologic conditions





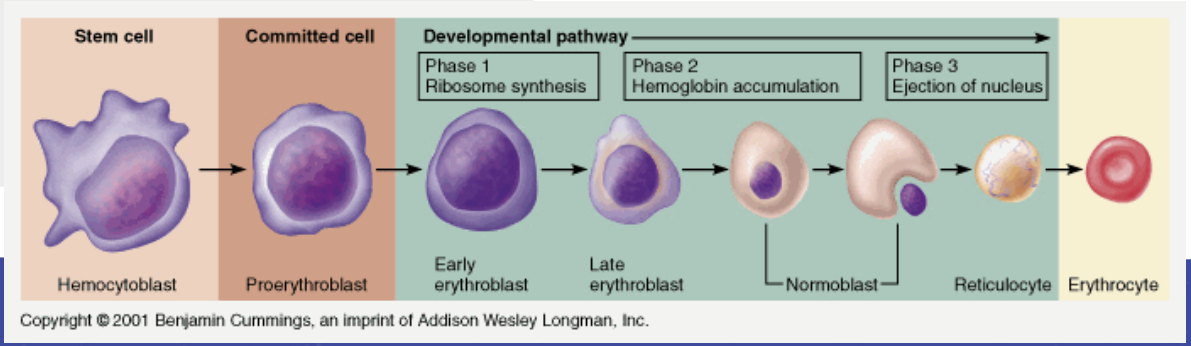
Chapter 23:

Disorders of Red Blood Cells



Red Blood Cell

- Erythroblasts are continuously being formed from stem cells in the bone marrow.
- Mature RBCs have no nucleus and are a biconcave shape



Function of Red Blood Cells

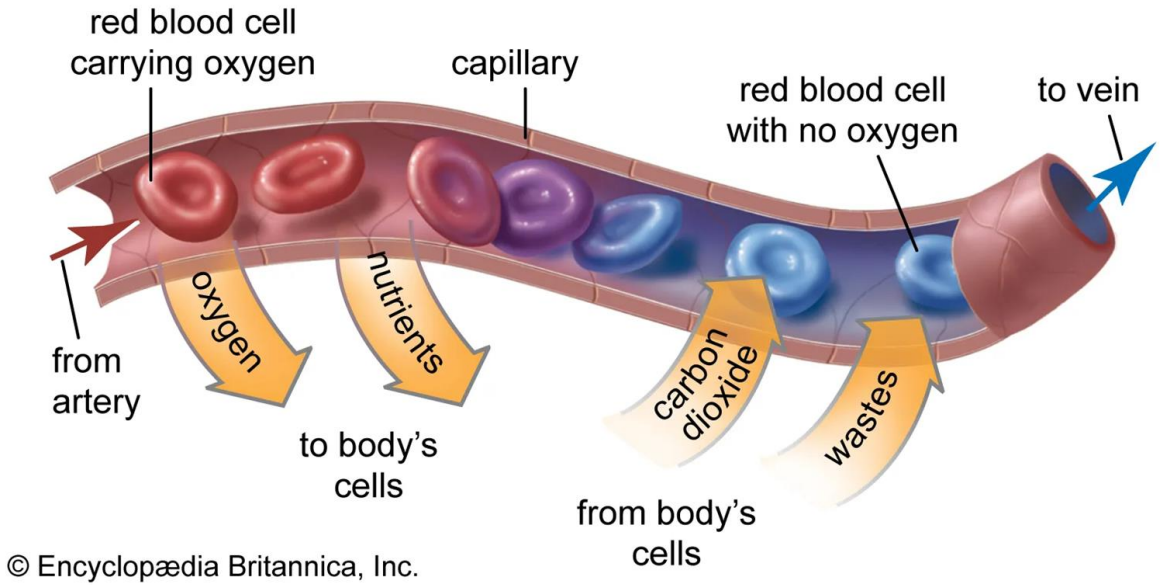
Transportation of oxygen from the lungs to the tissues

Hemoglobin binds carbon dioxide and carries it from the tissues to the lungs to be exhaled

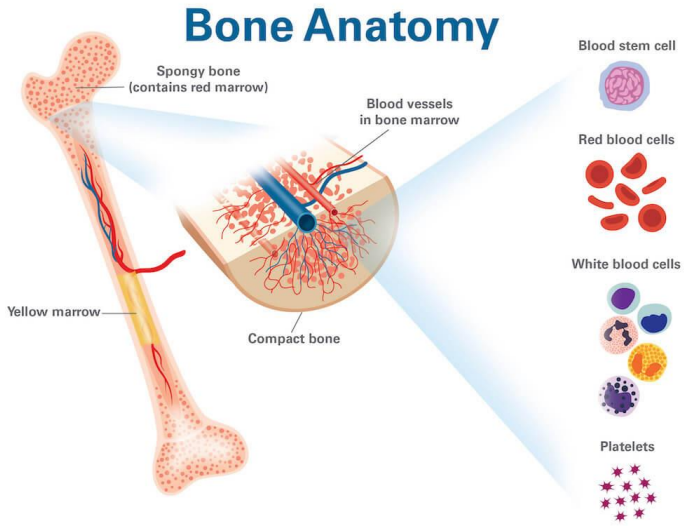
The hemoglobin molecule is composed of two pairs of structurally different polypeptide chains, alpha and beta

Each of the four polypeptide chains consists of a globin and a heme unit which bing to O2 or CO2

Function Continued



Erythropoiesis



Red cells are produced in the red bone marrow after birth.

Until 5 years of age, almost all bones produce red cells to meet growth needs; after 5 years, bone marrow activity gradually declines.

After 20 years, red cell production takes place mainly in the vertebrae, sternum, ribs, and pelvis.

With this reduction in activity, the red bone marrow is replaced with fatty yellow bone marrow.

Hemoglobin Stucture

Globin chains

α

β

Heme

Iron

Porphyrin

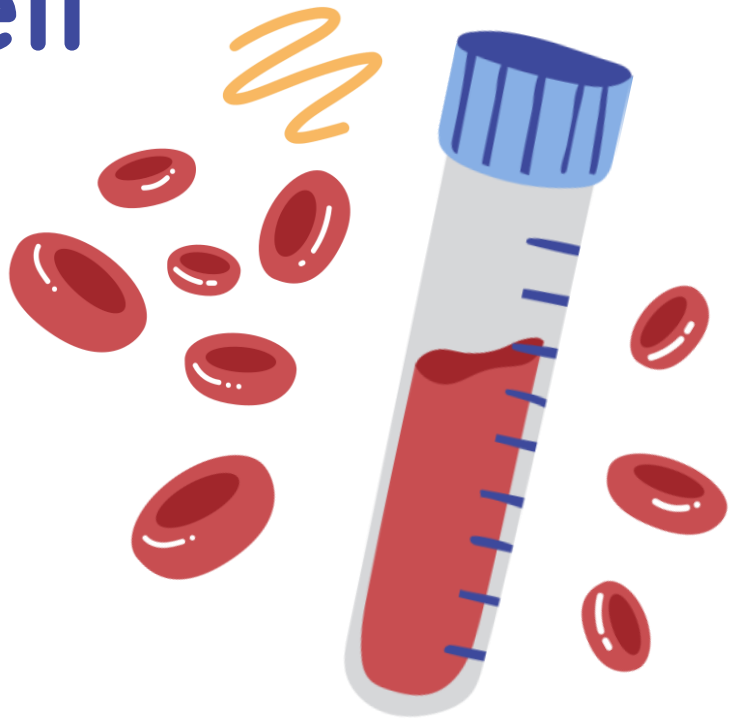
Oxygen delivery to tissues

Carbon Dioxide to lungs

Red Blood Cell Destruction

- The red blood cell has a life span of approximately 120 days.
  - It is broken down in the spleen.
  - The degradation products (iron and amino acids) are recycled.

- The heme molecule is converted to bilirubin and transported to the liver.
  - It is removed and rendered water soluble for elimination in the bile.



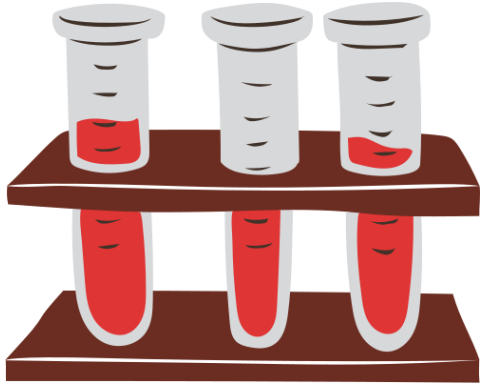
Bilirubin

- The heme unit is converted to unconjugated bilirubin in the spleen.
    - This is insoluble in the plasma which leads to toxicity
    - Must go to the liver for further processing
- Removed from the blood by the liver and conjugated with glucuronide to render it water soluble to be eliminated in bile
- Hyperbilirubinemia (Jaundice)

Laboratory Tests for Red Blood Cells

- Red blood cell count (RBC)
  - Measures the total number of red blood cells in 1 mm of blood
- Percentage of reticulocytes (normally approximately 1%)
  - Provides an index of the rate of red cell production

Laboratory Tests for Red Blood Cells



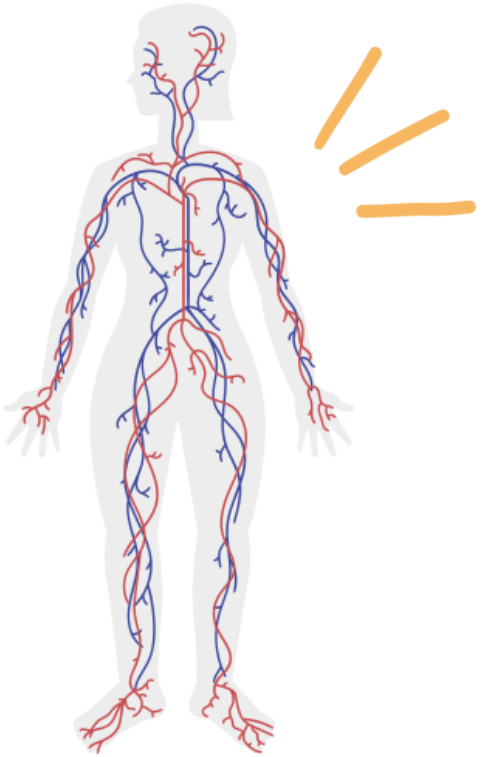
- Hemoglobin (grams per 100 mL of blood)
  - Measures the hemoglobin content of the blood
- Hematocrit
  - Measures the volume of red cell mass in 100 mL of plasma volume

Transfusion Therapy

- Provides the means for replacement of red blood cells and other blood components

- Four major ABO blood types are determined by the presence or absence of two red cell antigens: A and B.
  - The presence of D antigen determines the Rh-positive type.

- The absence of the D antigen determines the Rh-negative type.



Red Blood Cell Components Used in Transfusion Therapy

- Whole blood
- Red blood cells
- Leukocyte-reduced blood cells
- Washed red blood cells
- Frozen red blood cells



Signs and Symptoms of Transfusion Reactions

- Sensation of heat along the vein where the blood is being infused
- Flushing of the face
- Urticaria, headache, pain in the lumbar area
- Chills, fever, constricting pain in the chest
- Cramping pain in the abdomen
- Nausea, vomiting
- Tachycardia, hypotension, and dyspnea

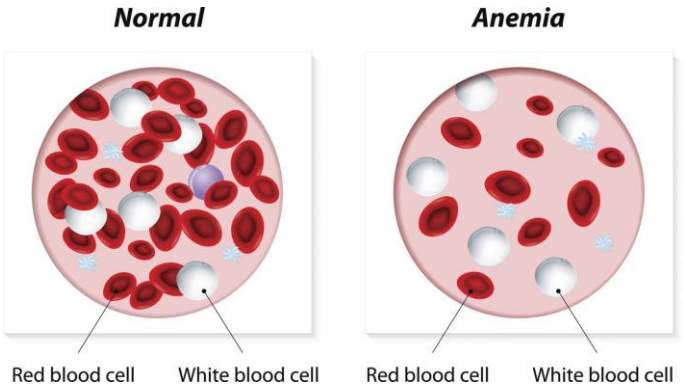
Anemia

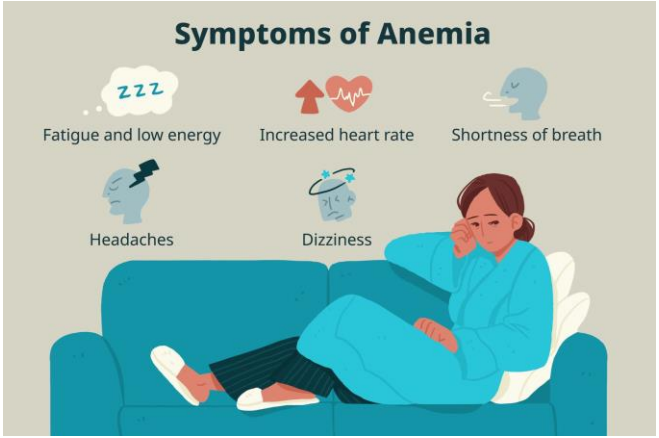
An abnormally low number of circulating red blood cells or level of hemoglobin, or both

- Results in diminished oxygen-carrying capacity

4 primary causes

1. Excessive loss of RBCs from bleeding
2. Destruction of RBCs
3. Defective RBC production
4. Inadequate RBC production due to bone marrow failure

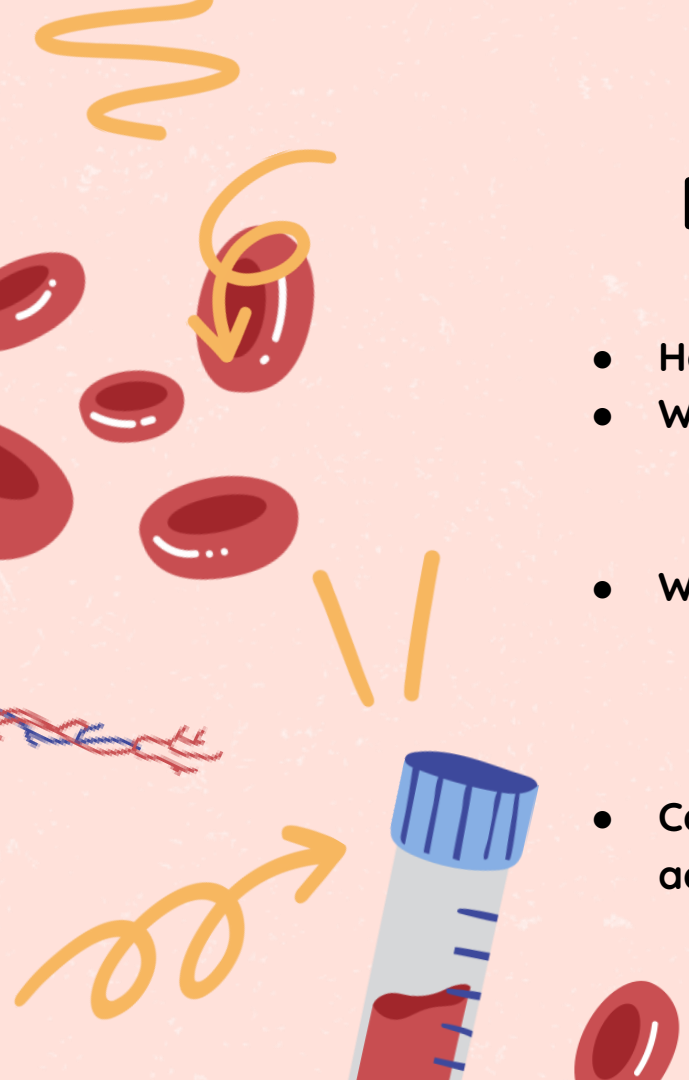




Manifestations of Anemia

- Impaired oxygen transport which can result in many issues such as fatigue, dizziness, SOB, headaches, paleness

- In severe cases can cause tachycardia and other heart issues

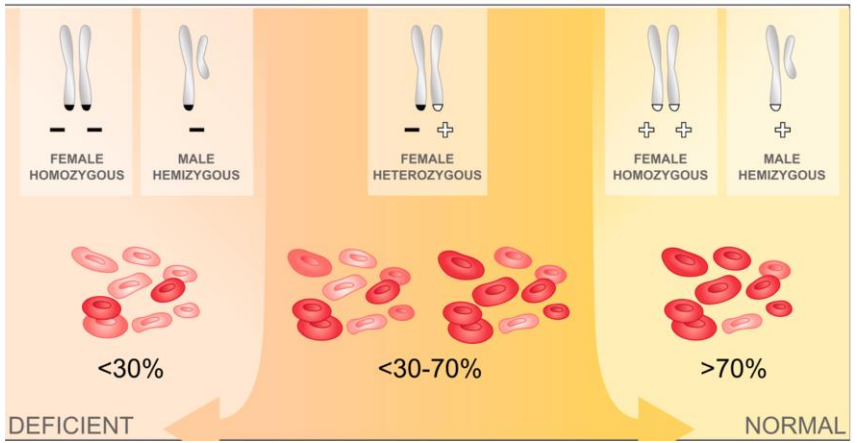


Hemolytic Anemias

- Hemolytic anemia is the premature destruction of RBCs
- Where does it happen?
  - Intravascular- inside the blood vessels
  - Extravascular- in spleen/liver
- What causes it?
  - Intrinsic- RBCs are defective
  - Extrinsic-RBCs are normal but get destroyed by external factors
- Can be inherited (sickle cell anemia, G6PD, thalassemias) or acquired (infections, toxins, chemicals)

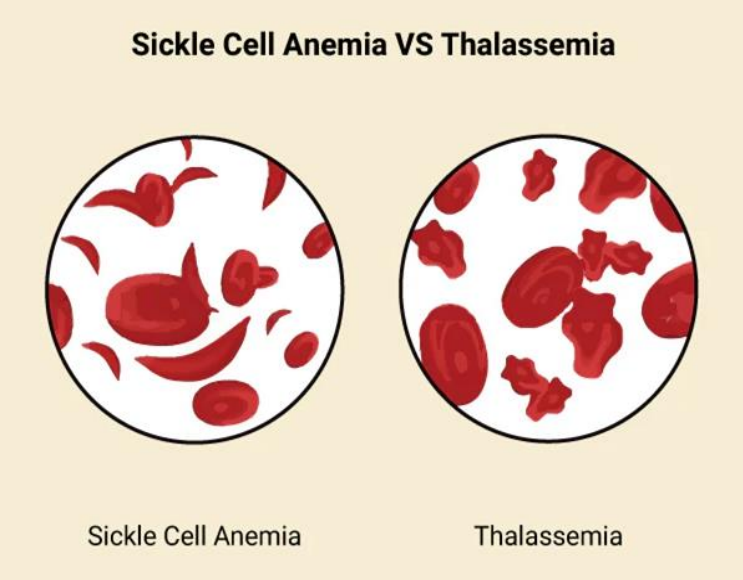
G6PD

- G6PD is an enzyme that helps protect RBCs
- Deficiency of this glucose-6-phosphate dehydrogenase is a hereditary genetic disorder
- X linked recessive disorder so more common in males
- Triggered by infections, drugs, and fava beans

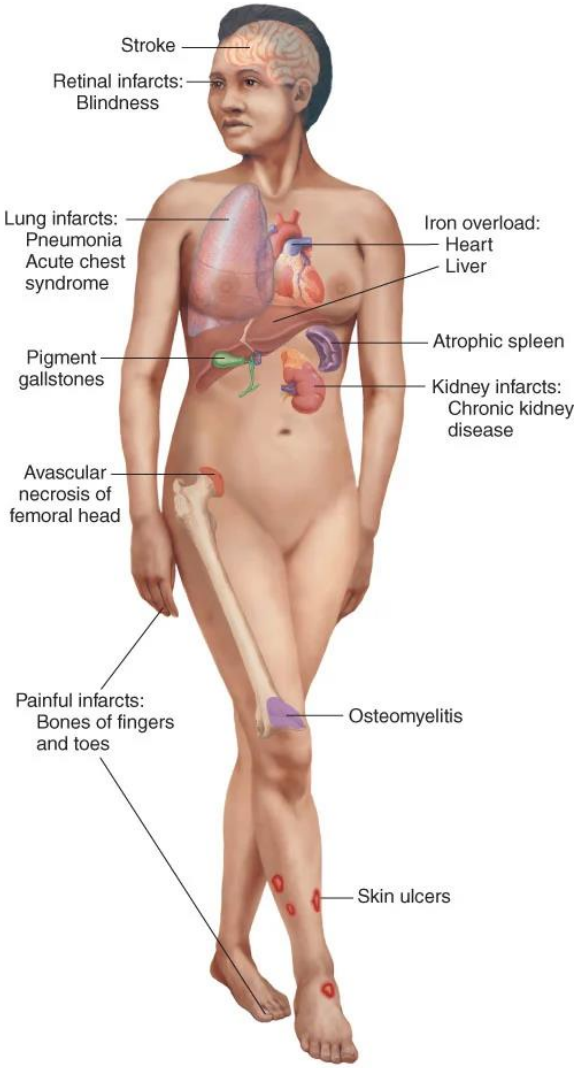


Sickle Cell Disease and Thalassemias

- Sickle cell disease is an inherited disorder in which an abnormal hemoglobin leads to chronic hemolytic anemia, pain, and organ failure.
- The thalassemias are a group of inherited disorders of hemoglobin synthesis leading to decreased synthesis of either the alpha or beta globin chains of HbA.



Clinical Manifestations of Sickle Cell disease



Anemias of Deficient Red Cell Production

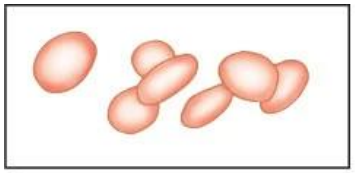
Iron deficiency anemia

Megaloblastic anemia

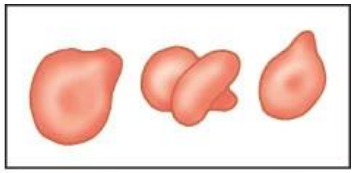
Aplastic anemia

Chronic disease anemia

Iron-Deficiency Anemia



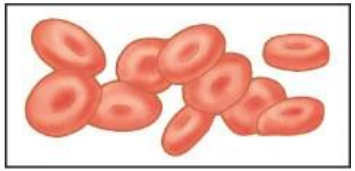
A Iron deficiency anemia



B Megaloblastic anemia



C Sickle cell disease

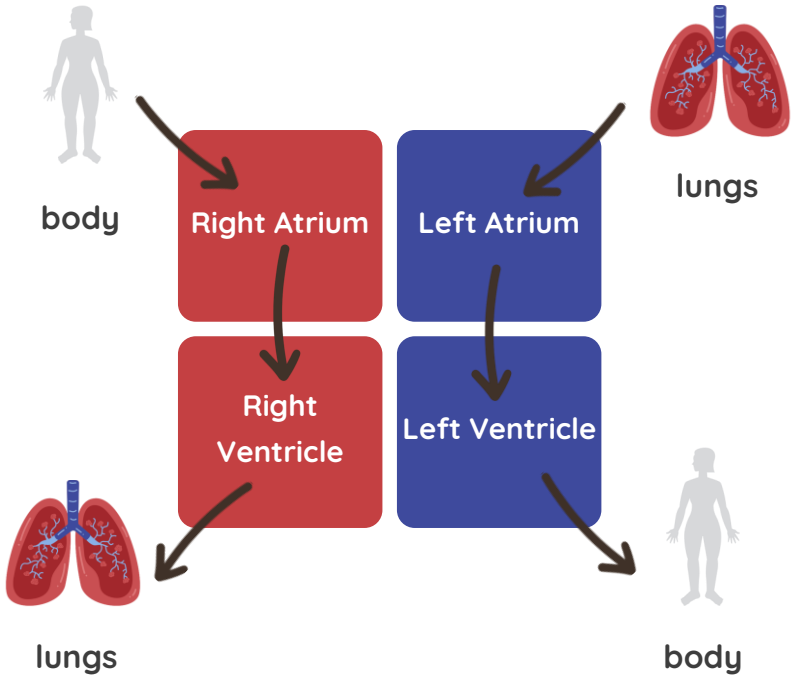


D Normal

- Iron deficiency is a common cause of anemia affecting people of all ages. It can occur by dietary insufficiencies of iron
- Because iron is a component of heme, a deficiency leads to decreased hemoglobin synthesis and consequent impairment of oxygen delivery.
- Treatment includes iron supplements

Vitamin B12 Deficiency

- Vitamin B12 deficiency causes megaloblastic anemia.
- When vitamin B12 is deficient, the red cells that are produced are abnormally large because of excess cytoplasmic growth and structural proteins.

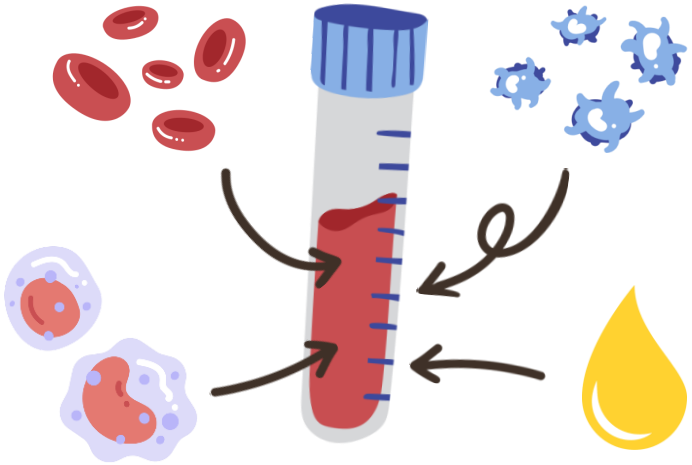


Causes of Aplastic Anemia

- causes are exposure to high doses of radiation, chemicals, and toxins that suppress hematopoiesis directly or through immune mechanisms.
- Results in a reduction of RBC, WBC, and platelets
- Life threatening

Polycythemia #1

A condition in which the red blood cell mass is increased.



Relative polycythemia: results from a loss of vascular fluid and is corrected by replacing the fluid

Polycythemia #2

Primary polycythemia: a proliferative disease of the bone marrow with an absolute increase in total red blood cell mass accompanied by elevated white cell and platelet counts

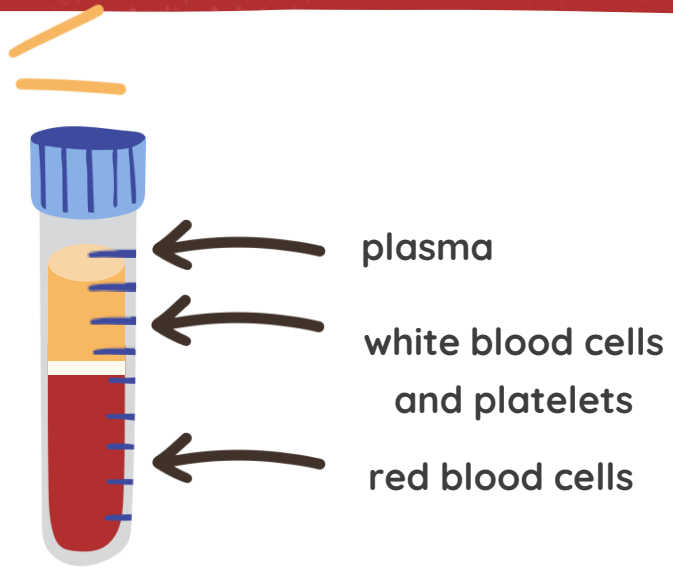
Secondary polycythemia: results from increased erythropoietin levels caused by hypoxic conditions such as chronic heart and lung disease

Polycythemia Manifestations

- Variable and related to an increase in RBCs, hemoglobin level, and hematocrit with increased blood volume and viscosity
  - Splenomegaly
  - Depletion of iron
  - Disrupted cardiac output
  - Hypertension
  - Decreased cerebral blood flow
  - Venous stasis
  - Thromboembolism and hemorrhage

Neonatal Blood

- Hemoglobin concentrations at birth are high, reflecting the high synthetic activity in utero to provide adequate oxygen delivery.
  - HbF versus HbA
    - Affinity of F versus A
- Hyperbilirubinemia
  - Unconjugated versus conjugated
  - Phototherapy
- Hemolytic disease



Blood is divided into its components, with the heaviest parts at the bottom.

Aging and Red Blood Cells

- Age-associated decline in the hematopoietic reserve
  - Reduction in hematopoietic progenitors
- Reduced production of hematopoietic growth factors
- Inhibition of erythropoietin
- Inflammatory cytokines interfere with erythropoietin interaction with its receptors.



Aging and Red Blood Cells

Anemia

Hb decreases with age.

RBC production decreases with age.

Complete blood count

Peripheral blood smear and a reticulocyte count and index

Studies to rule out comorbid conditions such as malignancy, gastrointestinal conditions that cause bleeding, and pernicious anemia

QUESTION 1

True or False

Manifestations of anemia are dependent upon the causative factor in the initiation of the disease

Answer to #1

False

Regardless of the cause, the main effect of anemia is hypoxia

Question #2

Which of the following diseases may be directly related to dietary insufficiencies?

a) G6PD
b) Sickle cell anemia
c) Polycythemia
d) Iron deficiency anemia

Answer to #2

D) Iron deficiency anemia

Can be caused by decreased iron intake from food

Question #3

Which Vitamin is most closely associated with the physiology of hemostasis?

a) Vitamin A
b) Vitamin K
c) Vitamin C
d) Vitamin D

Answer to #3

b) Vitamin K

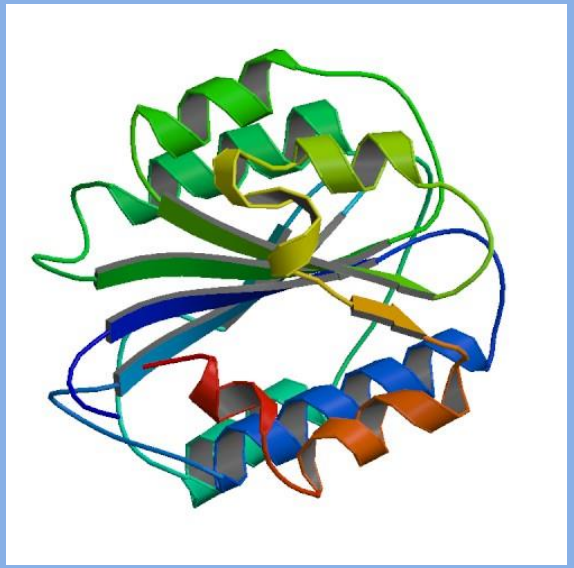
Question #4

Which protein molecule is required to form a platelet plug?

a) Thromboxane a2
b) Lipoprotein
c) Von WIllebrand factor
d) Plasminogen

Answer to #4

c) Von Willebrand Factor



Thank you!



